

WASP-52b

R-0468

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-52_250831 · created 2026-07-20T00:55:04+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460918.8056 +/- 0.0091 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.216 +/- 0.063
Transit depth (Rp/Rs)^2	4.68 +/- 2.71 [%]
Orbital Inclination (inc)	86.42 +/- 2.73
Airmass coefficient 1 (a1)	2.25 +/- 1.53
Airmass coefficient 2 (a2)	-0.32 +/- 1.16
Scatter in the residuals of the lightcurve fit is	68.43 %
Best Comparison Star	None
Optimal Aperture	1.82
Optimal Annulus	4.75
Transit Duration (day)	0.08 +/- 0.017

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.216 ± 0.063 vs 0.1646 (deviation 0.82σ)
Duration fit vs published	0.08 d vs 0.0758 d (deviation 0.25σ)
Mid-time O–C	21.8 min
Depth SNR	3.4
Residual scatter	68.43 %

Light curve

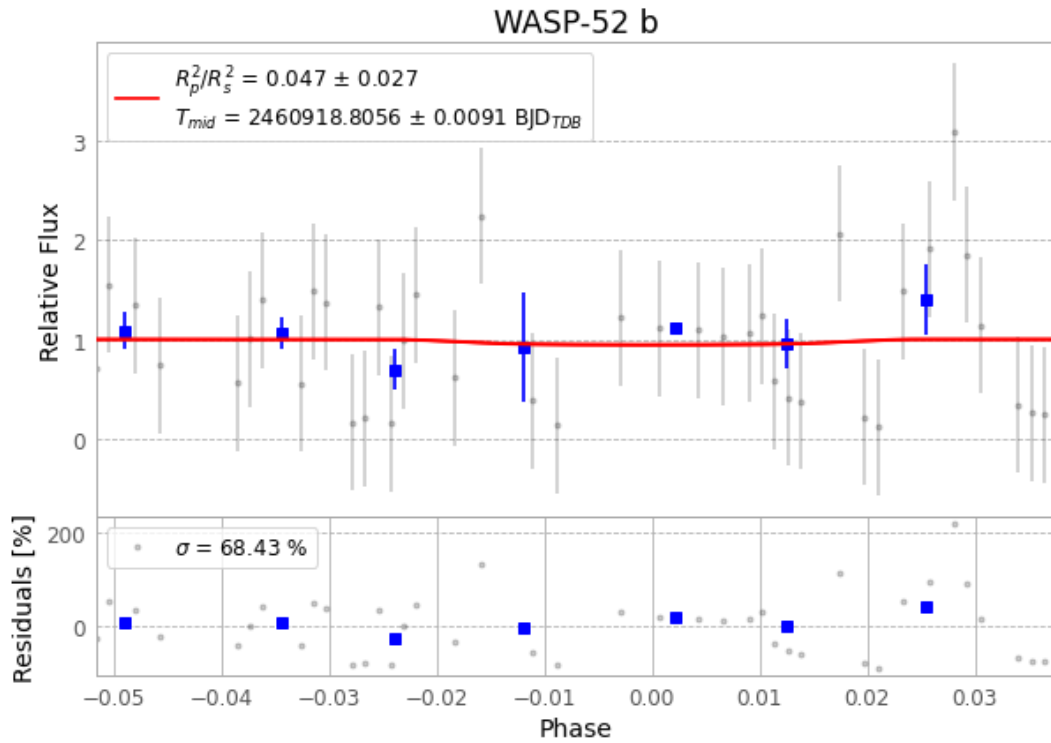


Figure 1 — FinalLightCurve_WASP-52 b_2025-08-30.png (SHA-256 fccac58659a8cde9...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [275, 390], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 935b2a01e9312a7c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

77 FITS frames (51 MB), WASP-52250831045715.FITS → WASP-52250831084515.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-250831.log (d779aa0e7399...), FinalLightCurve_WASP-52 b_2025-08-30.png (fccac58659a8...), FinalLightCurve_WASP-52 b_2025-08-30.csv (b070229f2de1...)

Compute resources

Wall time 100.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-20 00:55Z.